

Deepak Devapooja | Healthcare Data Analyst

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SUMMARY

Healthcare Data Analyst with nearly 6 years of experience in transforming large-scale clinical, claims, and operational datasets into actionable insights. Proficient in **SQL, Python, and R** for **ETL pipeline automation, predictive analytics, and data modeling**, enabling faster and more accurate healthcare decision-making. Skilled in **Tableau and Power BI** for developing interactive dashboards that enhance visibility into **population health, cost efficiency, and regulatory compliance**. Experienced in **EHR/EMR systems** (Epic, Cerner, Allscripts) and healthcare data standards including **HL7, FHIR, ICD-10, and CPT**, delivering **scalable, automated, and compliant solutions** that improve **data quality, governance, and operational performance** across healthcare organizations.

TECHNICAL SKILLS

Healthcare Data Analytics & Reporting: SQL (complex queries, joins, CTEs, stored procedures), Python (pandas, NumPy, SciPy, Matplotlib, Seaborn, scikit-learn), R (dplyr, ggplot2, caret), Excel (Pivot Tables, Power Query, VBA), ETL development, data wrangling, healthcare claims analysis, statistical modeling, hypothesis testing

Healthcare Systems & Data Standards: EHR/EMR systems (Epic, Cerner, Allscripts), HL7, FHIR, ICD-10, CPT, LOINC, DRG, Medicare and Medicaid datasets, claims adjudication, revenue cycle management (RCM), healthcare data warehousing, automated data pipelines

Data Visualization & Business Intelligence: Tableau, Power BI, QlikView, Looker, interactive dashboard design, KPI and metric reporting (HEDIS, STAR ratings, readmission, LOS, utilization, AR aging), data storytelling for clinical and financial insights

Predictive & Advanced Healthcare Analytics: Predictive risk scoring, disease progression forecasting, readmission and utilization prediction, survival analysis, A/B testing for clinical interventions, cost and reimbursement modeling, fraud and anomaly detection, machine learning (regression, random forest, clustering)

Data Governance & Compliance: HIPAA and GDPR compliance, PHI/PII security, data quality validation, data lineage documentation, audit trail implementation, role-based access control (IAM policies), regulatory reporting (CMS, NCQA, HEDIS)

Statistical & Quantitative Techniques: Descriptive and inferential statistics, ANOVA, confidence intervals, survival and time-series analysis, clinical outcome validation, population health and quality studies, variance and trend analysis

Collaboration & Project Management: Agile/Scrum methodology, Jira, Confluence, ServiceNow, BRD and SRS documentation, process flow design, cross-functional collaboration with clinical, finance, and compliance teams, stakeholder reporting and presentation

PROFESSIONAL EXPERIENCE

Healthcare Data Analyst | Cigna Healthcare | USA

Dec 2024 – Current

- Examined complex **healthcare claims, patient demographics, and pharmaceutical data** using **Python, R, and SQL** to identify **cost drivers, utilization patterns, and care gaps**, improving clinical outcomes by **14%** and reducing overall healthcare costs by **9%**.
- Developed and automated **ETL workflows** with **SQL and Python** to extract, clean, and validate high-volume healthcare data from multiple systems, reducing manual reporting time by **30%** and ensuring **data accuracy above 98%**.
- Designed and optimized **interactive dashboards** in **Tableau and Power BI**, enabling leadership to monitor **patient outcomes, claims utilization, and financial performance**, accelerating decision-making cycles by **25%**.
- Collaborated with **clinicians, data engineers, and business stakeholders** to translate complex healthcare challenges into analytical solutions, aligning with **compliance, governance, and organizational objectives**.
- Performed **predictive modeling and statistical analysis** using **Python and R** to forecast claims trends, detect anomalies, and identify high-risk patient populations, supporting **preventive care and fraud detection initiatives**.
- Ensured robust **data governance and HIPAA compliance** by implementing quality checks and maintaining **data integrity** across all analytical processes.
- Conducted **claims and cost optimization analyses**, delivering actionable insights from **clinical, operational, and financial data** to drive **resource optimization, process improvements, and operational efficiency**.

Senior Analyst | R1 RCM | India

May 2020 - May 2023

- Managed **Accounts Receivable (AR) follow-ups** by tracking claim statuses and resolving unpaid or underpaid claims using **SQL, Excel, and Azure data tools**, ensuring **timely reimbursements** and reducing outstanding AR by **12%**, improving cash flow for healthcare clients.
- Led **Denial Management initiatives** by identifying root causes of claim denials, implementing corrective actions, and developing preventive strategies. Increased **denial resolution rate by 15%**, enhancing revenue recovery and operational efficiency.
- Ensured **accurate medical billing and coding** for US healthcare accounts using **ICD/CPT codes** and adhering to **Medicare/Medicaid guidelines**, optimizing **reimbursement timelines** and minimizing compliance risks.
- Developed **interactive dashboards and reports** using **Power BI and Azure analytics services**, monitoring key RCM metrics such as **denial rates, AR aging, and revenue trends**, enabling leadership to make **data-driven decisions** that enhanced operational performance.
- billing and denial data trends** using **Python and Excel**, recommending and implementing **process improvements** that streamlined workflows, reduced operating costs, and improved efficiency across the RCM cycle.
- Mentored **junior analysts** and cooperated with **cross-functional teams** to meet client requirements, strengthen **data governance**, and ensure compliance with **US healthcare regulations** and RCM best practices.
- Monitored **data quality and performance**, conducting regular audits to maintain **accuracy, integrity, and regulatory compliance**, ensuring reliable reporting and analytics for client operations.

- Collected, cleaned, and validated large healthcare datasets, including EHR/EMR and claims data, using SQL, Excel, and Azure Data Services. Ensured data accuracy and compliance with healthcare standards, reducing reporting errors by 15%.
- Analyzed clinical and operational datasets using Python and R, identifying trends, patterns, and actionable insights. Supported population health initiatives and strategic decision-making, contributing to a 10% improvement in patient care workflows.
- Developed interactive dashboards and reports using Power BI and Azure analytics tools, enhancing visibility of key healthcare metrics such as patient outcomes, readmission rates, and resource utilization. Enabled leadership to make data-driven decisions for clinical and operational efficiency.
- Assisted in database design, maintenance, and Azure-based data integration, troubleshooting coding errors and optimizing queries. Improved data processing efficiency for large-scale healthcare datasets and reporting pipelines.
- Cooperated with cross-functional teams, including clinical, operations, and IT teams, to streamline workflows, implement process improvements, and strengthen data governance in healthcare processes.
- Monitored data quality and carried out regular checks, maintaining high standards of accuracy and integrity. Ensured compliance with healthcare regulations and reliable reporting for clinical and operational decision-making.

PROJECTS

Patient Readmission Prediction using EHR Data

- Built a machine learning model using Logistic Regression and Random Forest in Python to predict 30-day hospital readmissions, leveraging anonymized EHR datasets with diagnosis codes, lab results, demographics, and comorbidities.
- Performed comprehensive data preprocessing and feature engineering on ICD-10, CPT, and LOINC codes, refining variable selection and improving model interpretability and clinical relevance.
- Optimized model performance through cross-validation, hyperparameter tuning, and class balancing techniques, achieving an overall prediction accuracy of 82% and strong precision-recall balance.
- Collaborated with clinical analysts and data engineers to integrate predictive outputs into care management workflows, improving early intervention efficiency and reducing readmission rates by 10% in pilot testing.

Healthcare Quality Metrics & Population Health Dashboard

- Designed and implemented an interactive Tableau dashboard to track HEDIS measures including breast cancer screening, diabetes HbA1c control, and flu vaccination rates, improving visibility into population health performance across regions.
- Executed population segmentation analysis by age, region, and chronic condition prevalence, enabling teams to identify high-risk cohorts and launch targeted preventive care initiatives.
- Delivered predictive insights revealing care gaps and performance variance, which guided focused outreach campaigns and decreased care gaps by 9% in underperforming population segments.

Cost & Utilization Analysis for Chronic Disease Management

- Executed time-series analysis of hospital claims for chronic disease patients including diabetes and hypertension using SQL and R, uncovering patterns in utilization rates, cost drivers, and treatment outcomes.
- Examined length of stay (LOS), readmission rates, and emergency visit frequency, identifying a \$2M annual savings potential and providing data-backed recommendations to optimize care pathways and reduce inefficiencies.
- Built a systematized Python reporting pipeline to automate cost and utilization reports, reducing manual reporting time by 40% and ensuring timely, accurate data delivery to leadership and clinical teams.
- Cooperated with finance, operations, and care management teams to evaluate high-cost patient cohorts, driving strategies that enhanced resource allocation and lowered overall treatment costs.
- Demonstrated findings through Tableau dashboards and executive reports, enabling data-driven budgeting and evidence-based policy decisions that refined both financial and patient care outcomes.

EDUCATION

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| Master’s in Information Technology & Management Webster University | May 2023 - May 2025 |
| Master’s in MBA Finance Holy Mother PG College | Jul 2016 - Aug 2018 |
| Bachelor’s in Commerce St. Josephs Degree & PG College | Jun 2012 - Apr 2015 |